

afterVideoLesson3.3.pdf



abcd__1234



Paralelismo



3º Grado en Ingeniería Informática



Facultad de Informática de Barcelona (Fib)
Universidad Politécnica de Catalunya



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 **Gemini 3**
Nuestro modelo más preciso

Sintetiza horas de
Investigación en minutos

Necesito estudiar a fondo el comportamiento de la fotosíntesis según el tipo de planta y el entorno



 Deep Research x



 Fuentes v

 Archivos

Estudia con Gemini

Ya has abierto los apuntes,
te mereces ese descanso.

También te mereces que no te cobren
por tener una cuenta. **Cositas.**

Ven a la **Cuenta NoCuenta**

Saber más



Qualificació 5,00 sobre 5,00 (100%)

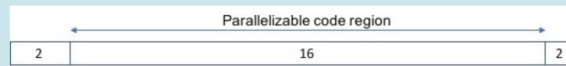
Pregunta 1

Completa

Puntuació 1,00
sobre 1,00

Marca la
pregunta

Given the following execution timing diagram of a sequential application:



In the timeline there are three computation bursts, the first with a duration of 2 time units, the second with a duration of 16 time units and the last one also with a duration of 2 time units. Only the second execution burst can be parallelized (decomposed into parallel tasks), the other two can not be parallelized.

Which is the parallel fraction (ϕ) of the application?

Trieu-ne una:

- ☐ 0.25
- ☒ 0.8
- ☐ 0.2

Pregunta 2

Completa

Puntuació 1,00
sobre 1,00

Marca la
pregunta

Which would be the speed-up that could be achieved using infinite processors (S_p when $p \rightarrow \infty$), assuming that the parallelizable region can be ideally decomposed into infinite tasks)?

Trieu-ne una:

- ☒ 5
- ☐ 1.33
- ☐ 1.25



do your thing

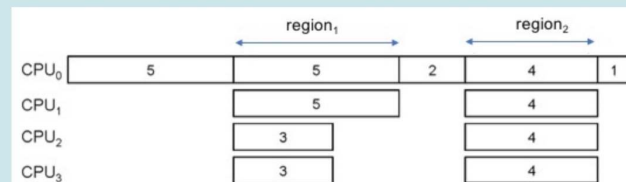
WUOLAH

Pregunta 3

Completa

Puntuació 1,00
sobre 1,00 Marca la
pregunta

Given the following execution timing diagram of an application executed on 4 processors (CPU₀ to CPU₃):



In the timeline there are two parallelizable regions: region₁ and region₂. In region₁ CPU₀ and CPU₁ execute computation bursts that last 5 time units, while CPU₂ and CPU₃ execute bursts that only last 3 time units (load unbalanced). In region₂ all processors execute a computation burst that lasts 4 time units (load balanced). Region₁ is preceded by a non-parallelizable computation bursts that lasts 5 time units. In between region₁ and region₂ there is non-parallelizable computation burst that lasts 2 time units. Finally, region₂ is followed by a non-parallelizable task that only lasts 1 time unit.

Which speed-up is achieved in the execution with 4 processors (S_4)?

Trieu-ne una:

- ☐ 2.5
- ☒ 2.35
- ☐ 4

Pregunta 4

Completa

Puntuació 1,00
sobre 1,00 Marca la
pregunta

Which is the parallel fraction (ϕ) of the application?

Trieu-ne una:

- ☐ 0.52
- ☐ 0.2
- ☒ 0.8

Pregunta 5

Completa

Puntuació 1,00
sobre 1,00 Marca la
pregunta

Which would be the ideal speed-up that could be achieved when using infinite processors (S_p when $p \rightarrow \infty$), assuming that the parallel regions can be decomposed into the infinite)?

Trieu-ne una:

- ☐ 2.08
- ☐ 1.25
- ☒ 5